React JS

React Router

What is React Router ?

- **React Router** is a **standard library used for routing** (navigation) in **React applications.**
- It helps you build **Single Page Applications (SPA)** by allowing navigation between different pages **without reloading the browser.**

Why React Router is Needed ?

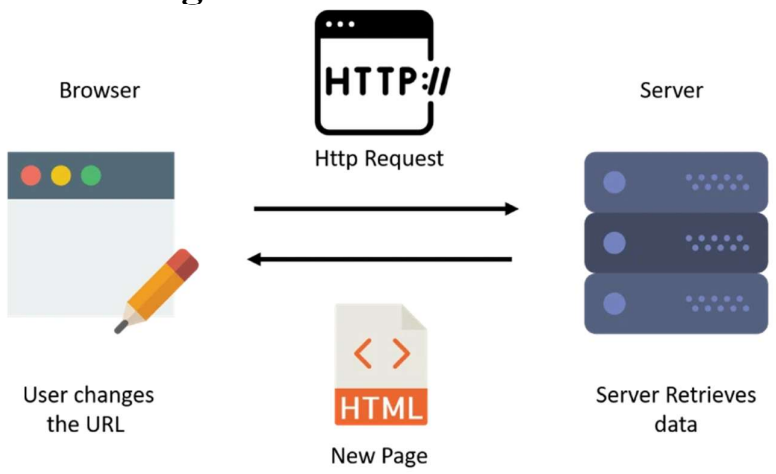
In React:

- By default, there is **no page navigation**
- Everything loads on a single page

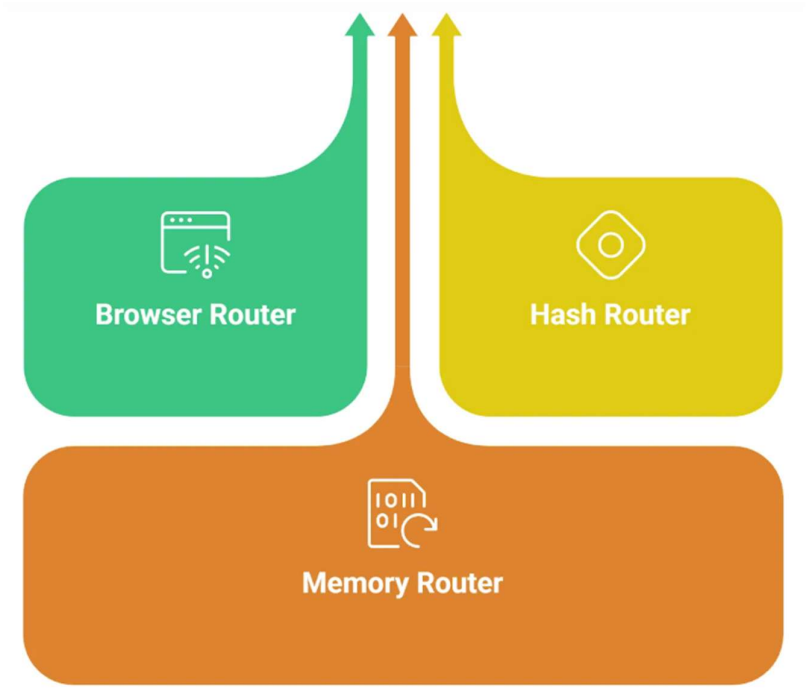
React Router:

- Maps **URLs → Components**
- Changes the **view** when the URL changes
- Keeps the app **fast and smooth**

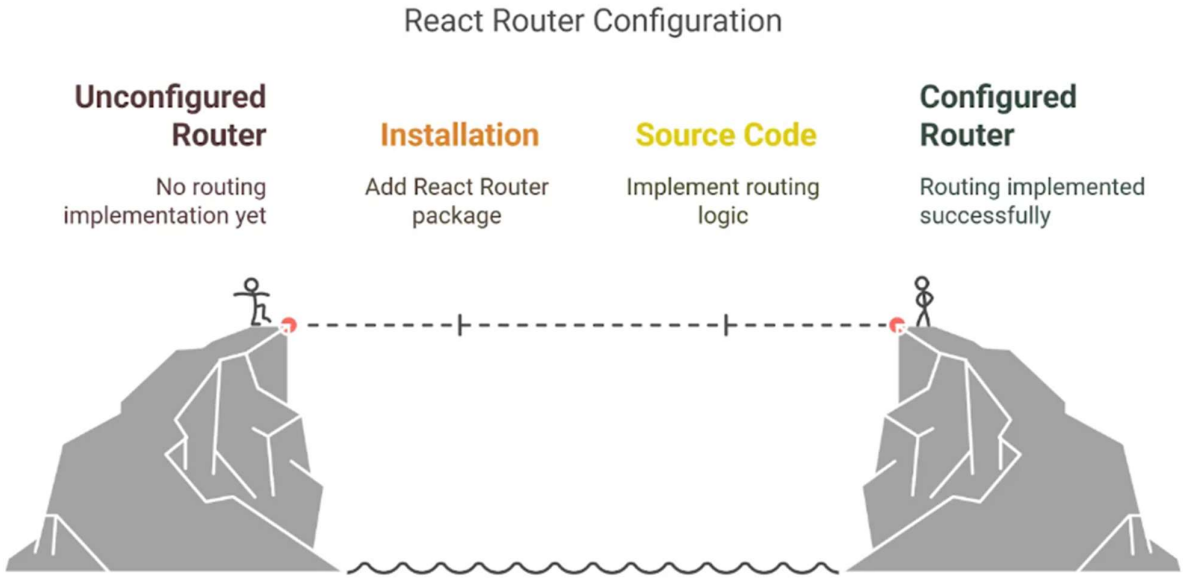
Process of Routing :-



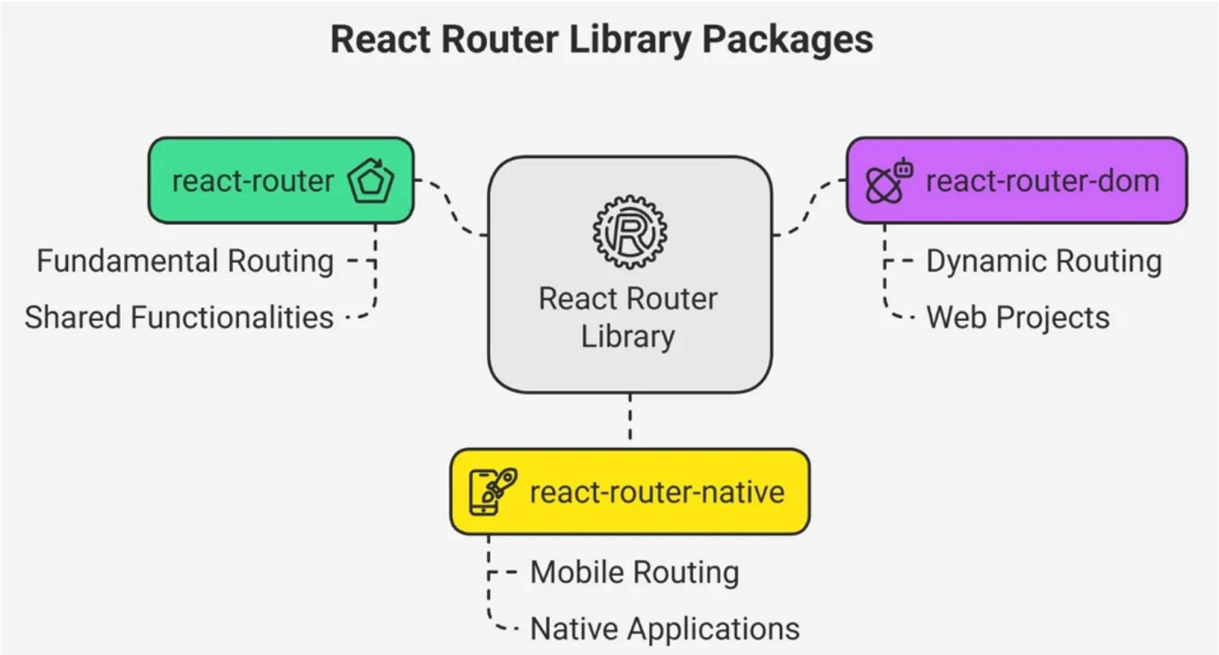
Types of React Routers :-



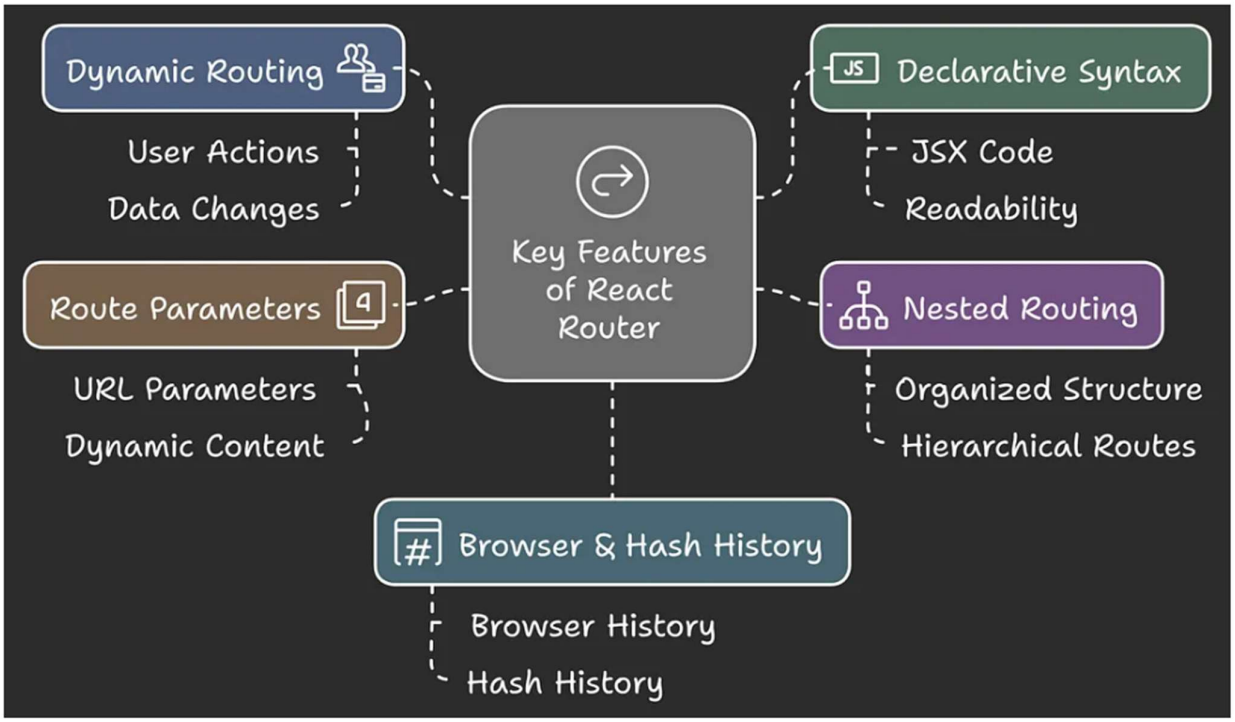
How Routing Works / Routing Mechanism :-



Various Packages in React Router Library :-



Features of React Router :-



Implementation :-

```jsx
import { BrowserRouter, Routes, Route } from "react-router-dom";

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/contact" element={<Contact />} />
      </Routes>
    </BrowserRouter>
  );
}
```

URL Mapping :-

URL	Component
/	Home
/about	About
/contact	Contact

Key Components of React Router :-

Component	Purpose
BrowserRouter	Wraps the app for routing
Routes	Container for all routes
Route	Defines path & component
Link	Navigation without reload
useParams	Get URL parameters
useNavigate	Programmatic navigation

Example: Navigation Link

```jsx
<Link to="/about">About</Link>
```

Real-Life Example :-

TV remote buttons -
Each button shows a different channel without turning off the TV